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Genetic risk of type 2 diabetes in populations of the African continent: A systematic review and meta-analyses

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abstract

Background: Type 2 diabetes (T2D) is growing faster in Africa than anywhere else, driven by the dual effects of genetic and environmental factors. We conducted a systematic review and meta-analyses of published studies on genetic markers of T2D in populations within Africa.

Methods: Multiple databases were searched for studies of genetic variants associated with T2D in populations living in Africa. Studies reporting on the association of a genetic marker with T2D or indicators of glycaemia were included. Data were extracted on study design and characteristics, genetic determinants, effect estimates of associations with T2D.

Findings: Overall, 100 polymorphisms in 57 genes have been investigated in relation with T2D in populations within Africa, in 60 studies. Almost all studies used the candidate gene approach, with >88% published during 2006–2014 and 70% (42/60) originating from Tunisia and Egypt. Polymorphisms in *ACE, AGRP, eNOS, GSTP1, HSP70-2, MC4R, MTHFR, PHLPP, POL1, TCF7L2, and TNF*-a gene were found to be associated with T2D, with overlapping effect on various cardiometabolic traits. The polymorphisms investigated in multiple studies mostly had consistent effects across studies, with only modest or no statistical heterogeneity. Effect

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sizes were modestly significant [e.g., odd ratio 1.49 (95%CI 1.33–1.66) for *TCF7L2* (rs7903146)]. Underpowered genome-wide studies revealed no diabetes risk loci specific to African populations.

Interpretation: Current evidence on the genetic markers of T2D in African populations mostly originate from North African countries, is overall scanty and largely insufficient to reliably inform the genetic architecture of T2D across Africa.

1. Introduction

The prevalence of diabetes has increased worldwide, including in Africa, where the greatest proportional increase (90%) in the number of diabetes cases is projected to occur by 2030 [1,2]. Type 2 diabetes (T2D) represents more than 90% of diabetes cases in Africa [1,2]. The surge in T2D in Africa has been primarily attributed to environmental factors due to nutrition transition and urbanization that bring about lifestyle changes; however, current understanding of the pathophysiology of T2D in populations within Africa remains elusive. From observations among other populations around the world, it is obvious that T2D has a strong genetic component, as evidenced by a high concordance rate (70%) in monozygotic twins [3] and the over 40% risk of having T2D for an offspring of a T2D patient [4]. Unlike simple monogenic diseases, the pathophysiology of T2D involves most of the time an interaction of several common genetic risk factors of low penetrance with environmental factors. Hitherto, the vast majority of T2D-related risk alleles have been identified through genome-wide association studies, mostly conducted in European Descendants and Asian populations [5]. These findings can hardly be extrapolated to other ethnic groups due to differences in biological traits, cultural practices, and lifestyle habits. Indeed, many common loci originally associated with T2D in Caucasians have not been replicated in non-European populations [6]. Furthermore, Africa is traditionally known to harbor more genetic variations than any other continent [7,8]. It is therefore important to understand the role of genes in populations within Africa as there may be specific genetic susceptibility to T2D in these groups. In addition, studying these populations through genome-wide association and fine mapping has the potential to provide valuable insights into the genetic architecture of type 2 diabetes and further explain geographical differences in prevalence.

This systematic review and meta-analysis synthesizes existing studies on genetic risk inferred by polymorphisms related to T2D among Africa-based populations; and explores the significance of the overall effects of the tested genes and heterogeneity in those effects across regions and major population groupings.

2. Methods

2.1. Data sources

We searched MEDLINE (via PubMed) and EMBASE for articles published until June 2014, without any date or language restriction. We used a combination of relevant search terms presented in appendix pp. 2–3. Two investigators (YYY and MGF) independently identified articles and sequentially (titles, abstracts, and then full texts) screened them for inclusion (Fig. 1). We also scanned the references lists of relevant publications, and identified their citations through the Web of Science. Any disagreements were solved by consensus or review by a third investigator (APK).

2.2. Study selection

We included genetic association studies conducted on African populations residing in the continent, that include diagnosed diabetes and/or markers of glucose homeostasis used as an outcome. Studies had to meet the following criteria: (i) original papers containing independent data, (ii) case-control or cohort, (iii) suffi data to calculate the odds ratio (OR) with a confidence interval (CI). We excluded duplicate publications; studies conducted exclusively on migrant populations of African descent living outside the continent. Fig. 1 shows the study selection process.

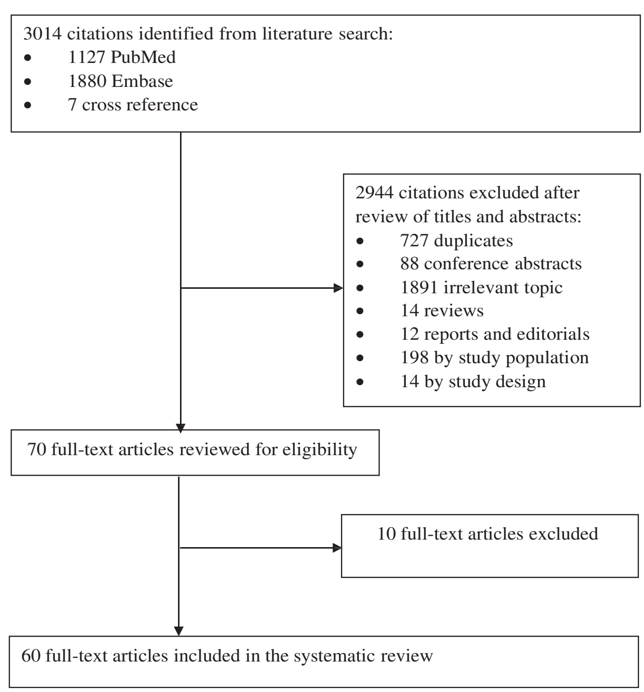

Fig. 1 – Selection process for studies included in the systematic review.

2.3. Data extraction, assessment, and synthesis

Two reviewers (YYY and MGF) independently extracted data from included studies, on settings, design, populations' characteristics, and measures of genetic association. There is currently no convincing evidence of specific study characteristics that can be used to assess the validity of genetic studies included in a systematic review. As recommended by Sagoo et al. [9], we assessed the presence of bias considering the following: case definition, population stratification, reporting of methods used (sample size of a study population, genotyping method and its reliability/accuracy, validation of results, statistical analyses).

For the polymorphisms investigated in multiple studies, we derived the pooled estimates of their association with diabetes risk across studies using a random effects model meta-analysis, implemented using STATA software (StataCorp, Texas, USA). We used the Cochran Q statistic which follows the chi square distribution [with k 1 degrees of freedom (df)] to assess the heterogeneity within and between included studies, with a p-value <0.05 indicating significant heterogeneity [10]. Heterogeneity in the measure of association across studies was further quantified with the I^2 statistic, with value <25% indicating low heterogeneity, 25–50% indicating moderate heterogeneity, 50–75% indicating high heterogeneity and >75% indicating extreme heterogeneity [11]. Publication bias was assessed using funnel plots and the Egger test of bias [12]. A funnel plot is a scatter plot of the effect estimates (log odds ratio in the current review) from individual studies against a measure of precision of those effect estimates (standard error of the log odds ratio in the current review). This measure of precision is often plotted on the vertical axis, and in a reverse scale that places large studies at the top and smaller studies at the bottom. Because estimates from larger studies are likely more precise, they will tend to cluster around the overall estimate (central line), while estimates from increasingly smaller and accordingly less precise studies will tend to be more scatter, but symmetrical about the central vertical line, giving an overall inverse funnel shape, when there is no publication bias. An asymmetrical funnel shape indicate a relationship between the effect size and precision in the studies at hands. This is often due to reporting bias (publication bias); although it could also be due to a systematic difference between larger and smaller studies, or to the presence of a subset of studies with a different mean effect size [13,14]. Because visual inspection of the symmetry of the funnel plot can be challenging, it is often supplemented by formal statistical tests (Egger test for instance).

3. Results

3.1. Literature searches

A total of 3014 published records were identified through searches. Titles and where necessary abstracts/summaries of these records were screened for relevance to the systematic review, and 2951 were excluded by disease, study population, and study design (mutation analysis, genetic marker/loci linkage studies, pharmacogenetics, and reviews). Fig. 2 shows a linear growth over time in the number of genetic studies on diabetes in Africa, with over 88% of eligible studies published between 2006 and 2014.

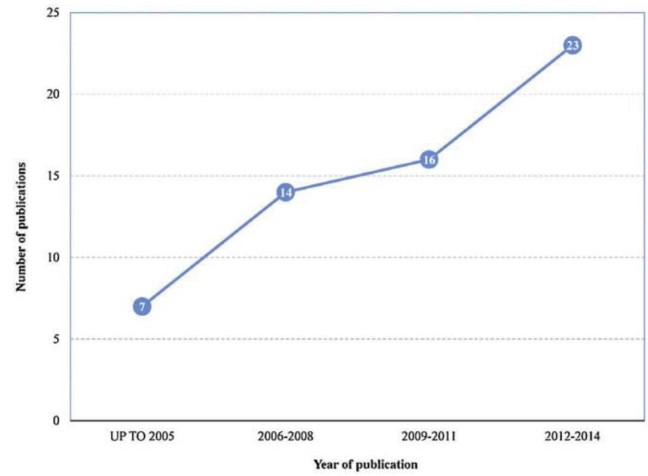

Fig. 2 – Time trend in genetic studies of diabetes in Africa.

3.2. Populations, genetic assessment and outcomes

The majority of included studies originated from two Northern African countries: Tunisia (n = 30) and Egypt (n = 11). The remaining studies originated from West Africa [Ghana (n = 6), Nigeria (n = 5), Mauritania (n = 2) and Senegal (n = 1)], Morocco (n = 5), South Africa (n = 4), and Democratic Republic of Congo and Mauritius (n = 1 each) (Table 1). The number of participants per study ranged from 30 to 2308, their age varied from 24 to 80 years, and the proportion of men from 13.7% to 61.2%. Diagnosis of diabetes was either based on WHO criteria (22 studies), ADA criteria (21 studies), or clinical diagnosis (three studies). Measures of glucose metabolism included fasting glycaemia (44 studies), post-prandial glycaemia (12 studies, including 2 h post-load glycaemia in four studies), and glycated hemoglobin (31 studies).

A total of 100 polymorphisms (including SNP, indels, and repeats) in 57 genes were investigated across studies. Appendix pp. 4–16 shows the distribution of the polymorphisms in various genes, including six in *ADIPOQ/ACDC* (rs16861194; rs17300539; rs266729; rs822396; rs2241767; rs1063538), five in *CAPN10* (UCSNP19; UCSNP-43; UCSNP-63; UCSNP-110; SNP-56), seven in the *KCNQ1* (rs231361; rs231359; rs151290; rs2237892; rs2283228; rs2237895; rs2237896), and five in *TCF7L2* (rs7903146; rs4506565; rs12243326; rs12255372; DG10S478). Two publications were on genome-wide scan of the Africa America Diabetes Mellitus (AADM) study population to search for type 2 diabetes susceptibility genes [15,16]. The nearest genetic markers on chromosomes regions with suggestion of linkage pick are shown in the appendix pp. 17.

3.3. Association of genetic markers with type 2 diabetes and related traits

As shown in Table S3 (appendix pp. 4–16), 59 SNPs in the *ACE, ADIPOQ/ACDC, Apo A-V, CAPN10, CDKAL1, DSEL, ENPP1, EXT2, HHEX, IFN-g, IGF2BP2, IL-6, IRS-1, IRS-2, KCNJ11, KCNQ1, MALT1,*

Table 1 – Characteristics of genetic association studies conducted in African populations.

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
Congolese (Democratic Republic of Congo)								
Bukavu	*FPN* Q248H	265	179/86	26–74	47.9	ADA criteria	PCR-RFLP	Katchunga et al. [51]
Egyptian								
Nile Delta region	*IL-6*-^{174} (G/C), *IL-10*–^{1082}(G/A)	167	69/98	40–78	44.9 (cases)	Medical history, HbA1C	Allele–specific PCR	Helaly et al. [52]
Not specified	*MTHFR* rs1801133 (C677T), *MTHFR* rs1801131 (A1298C)	120	60/60	37–53	na	Treated T2D, fasting glucose, HbA1C	PCR-RFLP	AbdRaboh et al. [53]
Not specified	IL-4-^{590} (C > T) and IL-13-^{1112} (C > T)	210	135/75	47–65	48 (cases)	WHO criteria,	Allele-specific PCR	Alsaid et al. [54]
Sinai area	*GSTM1* null, *GSTT1* null	200	100/100	40–56	49	ADA criteria, fasting glucose, HbA1c	Multiplex PCR	Amer et al., 2011 [55]
Not specified	rs1695 (*GSTP1* Ile105Val)	300	112/188	40–57	53.7	ADA criteria, fasting glucose, HbA1c	PCR-RFLP	Amer et al. [56]
Not specified	*TNF*-a rs1800629 (-308A), *IFN*- g rs2430561 (+A874T)	419	207/212	40–78	44.9 (cases)	Medical history, clinical examination, fasting and 2 h glucose, HbA1c	Amplification refractory mutation system-PCR	Elsaid et al., 2012 [57]
Not specified	+299(G > A) (rs3745367) and 420(C > G) (rs1862513) polymorphism in *RETN* gene	300	145/155	30–55	45.7	ADA criteria	PCR-restriction fragment length polymorphism	El-Shal et al. [58]
Not specified	*TGF*-b1 rs1982073 (T869C), rs1800471 (G915C)	197	99/98	39–56	16.2 (cases)	ADA criteria, fasting and 2 h glucose	Amplification refractory Mutation system- PCR	El-Sherbini et al. [59]
Not specified	*mEPHX1* rs1051740 Tyr113His, rs2234922 His139Arg	262	112/150	40–56	38.2	ADA Criteria, fasting glucose, HbA1c	PCR-RFLP	Ghatta and Amer [60]
Not specified	*ACE* rs4646994 (I/D), *MTHFR* rs1801133 (C677T), *MTHFR* rs1801131 (A1298C)	513	202/311	47–64	41.5	WHO criteria, fasting glucose, HbA1c	Allele-specific PCR and PCR-RFLP	Settin et al. [61]
Not specified	*ACE* rs4646994 (I/D)	45	24/21	32–62	51.1	WHO criteria,	PCR	Zarouk et al. [62]
Mauritania								
Moors, black Africans	*KCNJ11* rs5219 (E23K)	270	135/135	29–63	46.7	WHO criteria, Fasting glucose, HbA1c	TaqMan allelic discrimination	Abdelhamid et al. [63]

Table 1 (Continued)

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
Mauritius								
Creole/Indians	*GCK* (CA)$_n$ repeat	85/63	[40/45]/[31/32]	56–60/49–55	40/42.9	Medical history, WHO criteria	PCR and sequence	Chiu et al. [64]
Moroccan								
Not specified	C677T and A1298C Polymorphisms in *MTHFR*	514	282/232	46–68	34.8	WHO criteria, fasting glucose	PCR-RFLP	Benrahma et al. [65]
Arab	*CDKAL1*rs7756992, CNJ11rs5219, and *IGF2BP2* rs4402960	500	250/250	45–68	33.2	WHO criteria, fasting glucose	TaqMan allelic discrimination	Benrahma et al. [66]
Not specified	rs7903146 (C/T) in *TCF7L2*	910	504/406	43–69	30.4	ADA criteria, fasting glucose	TaqMan SNP genotyping	Cauchi et al. [25]
Not specified	*CDKN2A/2B* rs10811661, *CDKN2A/2B* rs564398, *CDKAL1* rs7754840, *CDKAL1* rs7756992, *CDKAL1* rs10946398, *IGFBP2* rs4402960, *IGFBP2* rs1470579, *EXT2* rs1113132, *EXT2* rs3740878, *EXT2* rs11037909, *EXT2* rs729287, *HHEX* rs1111875, *HHEX* rs7923837, *LOC646279* rs1256517, *SLC30A8* rs13266634, *MMP26* rs2499953, *KCTD12* rs2876711, *LDLR* rs6413504, *CAMTA1* rs1193179, *LOC387761* rs7480010, *NGN3* rs10823406, *CXCR4* rs932206	944	521/423	40	30.83	1997 ADA criteria	Taqman allelic discrimination	Cauchi et al. [67]
Arab-Berber	K121Q in *ENPP1*	915	503/412	40	30.6	ADA criteria, fasting glucose	Hybridisation probes on LightCycler	El Achhab et al. [68]
Senegalese								
Black population	L162V in *PPARa*	271	143/128	Not reported	na	Not reported	TaqMan allelic discrimination	Lopez-Sall et al. [69]

Table 1 (*Continued*)

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
South African								
Blacks	C-11377G and G-11391A in ACDC	453	227/226	26–69	13.7	WHO criteria, 2 h glucose	Real-time PCR	Olckers et al. [70]
Blacks	GGG → GGA at codon 235, Asp806Gly and Gly972Arg in *IRS-1*	30	10/20	25–50	50	Not reported	PCR-RFLP	Panz et al., 1997 [71]
Whites	GGG → GGA at codon 235, Asp806Gly and Gly972Arg in *IRS-1*	30	10/20	25–50	50	Not reported	PCR-RFLP	Panz et al. [71]
Blacks	C-11377G in *ADIPOQ*	453	227/226	26–69	13.7	WHO criteria,	Real-time PCR	Schwarz et al. [18]
Mixed-ancestry	*Pro12Ala* in *PPARG* and *Gly972Arg* in *IRS-1*	387	212/575	35–73	45.5	WHO criteria, Fasting glucose, HbA1c	Real-time PCR	Vergotine et al. [72]
Tunisian								
Not specified	*ACE* rs4646994 (I/D)	244	141/103	30–71	39.3	WHO criteria, Fasting and 2 h glucose, HbA1C	PCR	Arfa et al. [73]
Arab and Berber	*ACE* rs4646994 (I/D)	272	162/110	39–71	49.6	ADA criteria, fasting glucose	PCR	Baroudi et al. [74]
Arab and Berber	*IRS-2* rs1805097 (G1057D)	272	162/110	39–71	49.6	ADA criteria, fasting glucose	PCR-RFLP	Baroudi et al. [75]
Arab and Berber	*PstI* in INS, *NsiI* in INSR *BstnI* in IRS1	272	162/110	39–71	49.6	ADA criteria	PCR-RFLP	Baroudi et al. [76]
Arab and Berber	UCSNP19 polymorphism in *CAPN10*	272	162/110	39–71	49.6	ADA criteria	PCR and 6% polyacrylamide gel	Baroudi et al. [20]
Not specified	*PPARg-Pro12Ala* and *ENPP1*-K121Q	371	110/261	35–72	61.2	ADA criteria, fasting glucose	Light Cycler system and Light Typer system	Bouhaha et al. [77]
Not specified	*TNF-a-308G/A* and IL6 -174 G/C	528	228/300	35–72	57.2	ADA criteria, fasting plasma glucose	PCR-sequencing, Realtime PCR genotyping	Bouhaha et al. [31]
Not specified	rs7903146 (C/T) in *TCF7L2*	90	34/56	35–65	na	ADA criteria, fasting glucose	TaqMan allelic discrimination	Bouhaha et al. [26]
Not specified	SNP3 in (*Apo A-V*)	177	78/99	44–62	50.3	ADA criteria, fasting glucose, HbA1c	PCR-RFLP	Chaaba et al. [78]
Not specified	Glu298Asp, 4b/a, and 786T > C in eNOS	1665	917/748	48–70	47.7	WHO criteria, urinary albumin, HbA1c	PCR-RFLP	Ezzidi et al. [79]

Table 1 (*Continued*)

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
Arab	UCSNP-43 (rs3792267), UCSNP-19 (rs3842570), and UCSNP-63 (rs5030952) SNPs in *CAPN10*	1665	917/748	48–70	47.7	WHO criteria, fasting glucose, HbA1c	PCR-RFLP	Ezzidi et al. [21,80]
Arab	E23 K in *KCNJ11*/Kir6.2, K121Q in *ENPP1*, the -30G/A in GCK, rs7903146 in *TCF7L2* and rs7923837 in *HHEX*	1397	884/513	47–71	47.5	ADA criteria, fasting glucose, HbA1c	TaqMan Allelic discrimination	Ezzidi et al. [27]
Arab	UCSNP-43, UCSNP-19 and UCSNP-63 in *CAPN10*	1665	917/748	48–70	47.7	WHO criteria, fasting glucose, HbA1c	PCR-RFLP	Ezzidi et al. [21,80]
Not specified	Gly482Ser (rs8192678) in *PGC-1a*	889	487/402	31–59	41.8	WHO criteria, fasting glucose	PCR-RFLP	Jemaa et al. [81]
Not specified	863C/A in *TNF-a*	556	211/345	46–65	45.3	WHO criteria, fasting glucose	PCR-RFLP	Kallel et al. [32]
Not specified	UCSNP-43, UCSNP-19, UCSNP-110 and UCSNP-63 in *CAPN10*	428	222/206	49–74	45.3	WHO criteria, HbA1c	PCR-RFLP	Kifagi et al. [22]
Not specified	rs7903146 in *TCF7L2*, rs13266634 in SLC30A8, rs1111875 in *HHEX*, rs1113131 in *EXT2*, rs7480010 in LOC387761	734	331/403	49–75	45.2	WHO criteria, HbA1c	PCR-RFLP	Kifagi et al. [28]
Not specified	C667T polymorphism in *MTHFR*	206	86/120	43–65	44.2	ADA criteria, fasting glucose, HbA1c	PCR-RFLP	Koubaa et al. [82]
Not specified	XbaI, Enh2 and HpyCH4V	616	273/343	49–74	31.5	National Diabetes Data Group criteria, fasting glucose, HbA1c	PCR-RFLP	Makni et al. [83]
Not specified	I/D in *ACE*, C677T in *MTHFR*	231	115/116	38–65	41.1	WHO criteria, fasting glucose, HbA1c	PCR and PCR-RFLP	Mehri et al. [84]
Not specified	M235T in AGT, I/D in *ACE*, A1166C in AT1R,	289	114/175	38–65	37.7	ADA criteria, fasting glucose, HbA1c	PCR and PCR-RFLP	Mehri et al. [85]
Arab (Berbers excluded)	1082G/A, -819C/T, and 592C/A in IL-10 promoter	1665	917/748	47–70	47.7	Clinical and laboratory features, HbA1c	Allele-specific amplification-PCR	Mtiraoui et al. [86]

Table 1 (*Continued*)

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
Arabs	*ENPP1* rs1044498, *IGF2BP2* rs1470579, *KCNJ11* rs5219, *MLXIPL* rs7800944, *PPARG* rs1801282, *SLC30A8* rs13266634, *TCF7L2* rs7903146	2308	1470/838	49–71	45.9	WHO criteria, fasting glucose, HbA1c	Taqman allelic discrimination	Mtiraoui et al. [24]
Arabs	*ADIPOQ* promoter rs266730 rs16861194, rs17300539 rs266729, introns rs822395 rs822396, rs1501299 rs2241767, rs3774261 exons rs2241766 rs17366743, 3^{0}UTR rs1063537, rs1063538	1665	917/748	48–70	47.7	WHO criteria, fasting glucose, HbA1c	Taqman allelic discrimination, allele specific amplification (PCR-ASA) and PCR-RFLP	Mtiraoui et al. [17]
Not specified	26C/T and -90G/T in PRKAG2	51	28/23	49–72	Not reported	WHO criteria	direct sequencing	Nouira et al. [87]
Not specified	C677T in *MTHFR*	68	51/17	32–67	55.9	HbA1c	PCR	Oudi et al. [88]
Arab	rs231361, rs231359, rs151290, rs2237892, rs2283228, rs2237895, and rs2237896 in *KCNQ1*	1500	900/600	40–71	37.8 (cases)	WHO criteria, fasting and 2 h glucose, HbA1c	Real-time PCR allelic discrimination and PCR-RFLP	Turki et al. [89]
Arab	rs4506565, rs7903146, rs12243326, and rs12255372, in *TCF7L2*	1775	900/875	48–73	41.6	WHO criteria, fasting and 2 h glucose, HbA1c	allelic discrimination/ real-time PCR	Turki et al. [29]
Not specified	POL1-nearby variant rs488846, *MALT1*-nearby variant rs2874116, MC4R-nearby variant rs1942872, *PHLPP* rs9958800 and *DSEL*-nearby variant rs9966483 in the 18q region	1648	900/748	44–71	39.9	WHO criteria, fasting and 2 h glucose, HbA1c	PCR-RFLP	Turki et al. [90]
Not specified	308 *(TNF-a)* in *TNF-a*	554	280/274	28–63	na	HbA1c	PCR-RFLP	Zouari Bouassida et al. [33]
Not specified	1 267 *(HSP70-2)* in HSP70-2	541	267/274	28–63	na	HbA1c	PCR-RFLP	Zouari Bouassida et al. [33]

Table 1 (*Continued*)

Population	Polymorphism	Total sample	Number of cases/control	Age group (years)	% men	Diabetes diagnosis/ Measures of glucose homeostasis	Genotyping method	Reference
Not specified	*Pro12Ala* in PPARG	488	242/246	25–63	na	Fasting glucose, HbA1c	PCR-RFLP	Zouari Bouassida et al. [91]
West Africans								
Ghanaian	*TCF7L2* rs7903146 *KCNJ11* rs5219 *PPARG* rs1801282 *CAPN10* rs3842570, *CAPN10* rs3792267 *CAPN10* rs5030952	1052	675/377	24–68	24.7	Fasting glucose,	Specific primers and probes PCR on LightCycler 480 device, mutagenically separated PCR, PCR and PCR-RFLP	Danquah et al. [19]
Ghanaian (Accra and Kumasi) and Nigerian (Enugu, Ibadan and Lagos)	AGRP38C/T promoter SNP	538	381/157	24–80	41.3	ADA criteria, fasting and 2 h glucose	Pyrosequencing	Bonilla et al. [92]
Nigerian (Ibos and Yorubas)	SNP-43, SNP-56, and SNP-63 in *CAPN10*	294	197/97	36–65	46.6	ADA criteria, fasting and 2-h glucose	Pyrosequencing	Chen et al. [23]
Ghanaain (Akan-Ashante and Gaa)	SNP-43, SNP-56, and SNP-63 in *CAPN10*	201	150/51	38–63	36.9	ADA criteria, fasting and 2 h glucose	Pyrosequencing	Chen et al. [23]
Ghanaian (Akan and Gaa-Adangbe) and Nigerian (Yorubas and Igbo)	rs7903146, rs12255372 and DG10S478 in *TCF7L2*	1069	621/448	30–60		WHO criteria, fasting glucose	Sequencing	Helgason et al. [30]
Nigerian (Enugu, Ibadan and Lagos) and Ghanaian (Kumasi and Accra)	390 tri-nucleotide and tetra-nucleotide repeats	691	691/000	41–64	40.7	ADA criteria, fasting and 2 h glucose	S.A.G.E.-REALTEST based on likelihood method	Chen et al. [16]
Nigerian (Enugu, Ibadan and Lagos) and Ghanaian (Kumasi and Accra)	390 polymorphic markers	691	691/000	41–64	40.7	ADA criteria; fasting and 2 h glucose	GENEHUNTER-PLUS and ASM programs	Rotimi et al. [15]

ADA, American Diabetes Association; PCR, polymerase chain reaction; RFLP, restriction fragment length polymorphism; WHO, Word Health Organisation

*MLXIPL, MTHFR, PHLPP, PPAR*a, *PPARG, PPAR*g, *PRKAG2, SLC30A8, TCF7L2, TGF*-b1 and *TNF*-a genes, were examined and were not significantly associated with type 2 diabetes or measures of glycaemia across included studies (Table S2). In single studies some polymorphisms, namely the ones in the *ACE, AGRP, eNOS, GSTP1, HSP70-2, MC4R, MTHFR, PHLPP, POL1, TCF7L2* and *TNF*-a genes were found to be associated with type 2 diabetes in different ethnic groups (appendix pp. 4–16).

Of the six SNPs of the *ADIPOQ/ACDC* gene (rs16861194; rs17300539; rs266729; rs822396; rs2241767; rs1063538) investigated in two studies [17,18] including two types of population (one [rs266729] in Blacks South African and the rest in Arabs Tunisian), only two of these (rs1063538, rs2241767) were not associated with T2D in Arabs Tunisians. Among the five SNPs of the *CAPN10* gene (UCSNP19; UCSNP-43; UCSNP-63; UCSNP-110; SNP-56) investigated in five studies [19–23] across three populations (Ghanaian, Tunisian and Nigerian); four (UCSNP19, UCSNP-43, UCSNP-63 and SNP-56) were found to be associated with T2D. However, in the same Arabs population in Tunisia, Ezzidi and coworkers [21] found a significant association of UCSNP19 with diabetes risk, while Ouederni and coworkers [20] found no association.

Of the five SNPs of the *TCF7L2* gene (rs7903146; rs4506565; rs12243326; rs12255372; DG10S478) investigated in eight studies[19,24–30] across four population groups (Ghanaian, Tunisian, Morocco and Nigerians); only one (rs7903146) was associated with type 2 diabetes. Amidst the two SNPs (rs1800630 and rs1800629) of the *TNF*-a gene were investigated in four studies [31–33], in two non-specified populations (Tunisian and Egyptian), rs1800630 was found to be related to type 2 diabetes in a Tunisian population [32].

In the two genome-wide studies, significant linkage signal regions identified were 10q23 and 4p15, while few other regions had logarithm odds scores considered as suggestive evidence of linkage (appendix pp. 17) [15,16].

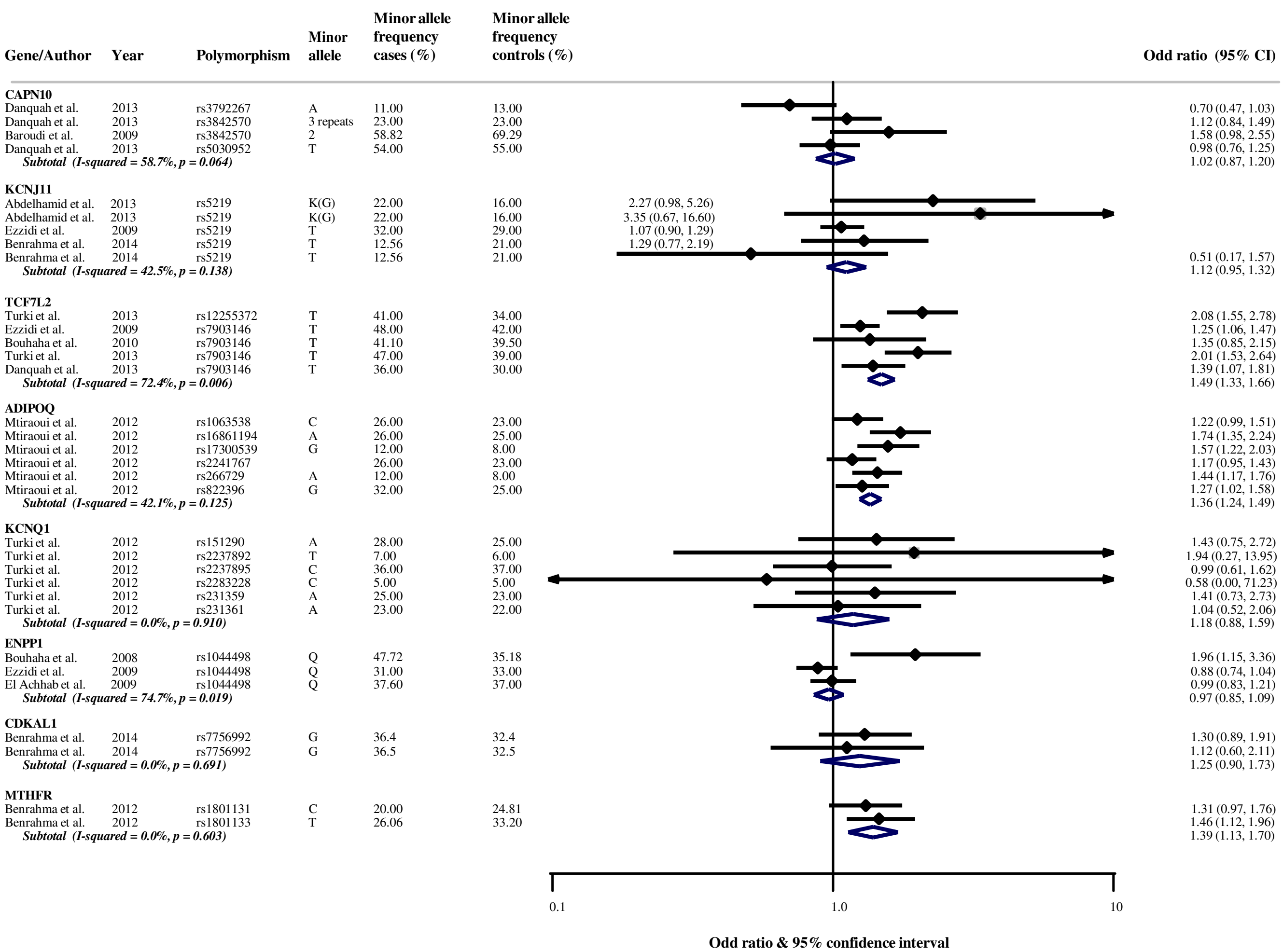

Fig. 3 – Forest plot for the effects of genes from additive models on the risk of diabetes across studies in Africa. For each included study, the solid diamond is for the effect size (odd ratio) of the association of the relevant genetic marker with diabetes risk, and the size of the diamond is proportional to the inverse variance of the effect estimate. The horizontal bars about the solid diamond represent the 95% confidence intervals. For each genetic marker, the bottom opened diamond is for the pooled effect size based on random effects models. For each study, the minor allele and its frequency in participants with diabetes (cases) and those without (controls) are shown as well as the odd ratio and the 95% confidence interval for the association of the genetic marker of interest with diabetes risk. For each genetic marker, measures of heterogeneity of the association with diabetes risk across included studies (*I*-squared and *p*-value) are shown.

3.4. Quantitative data synthesis

The pooled estimates of the effect of various genes are shown in Fig. 3 and appendix pp. 18–21. Across five studies that assessed the effects of the *TCF7L2* gene using additive models, the pooled odd ratio (OR) was 1.49 (95% confidence interval [CI]: 1.33–1.66; $I^2 = 42.5\%$, p-heterogeneity = 0.138). Equivalent figures were 1.41 (1.08–1.85; number of studies = 2; $I^2 = 0\%$, p-heterogeneity = 0.612) from recessive models, and 1.42 (1.34–1.51; number of studies = 20; $I^2 = 1.8\%$, p-heterogeneity = 0.435) from unspecified models. Therefore, regardless of the genetic model applied *TCF7L2* appeared to confer a 41%–45% higher odd of diabetes in African populations, with this effect being similar across studies and genetic models. Pooled ORs for the *MTHRF* gene were 1.39 (1.13–1.70; number of studies = 2; $I^2 = 0\%$, p-heterogeneity = 0.603) from additive models, 0.99 (0.83–1.18; number of studies = 4; $I^2 = 90.2\%$, p-heterogeneity < 0.001) from dominant models, 1.27 (0.91–1.77; number of studies = 4; $I^2 = 61.1\%$, p-heterogeneity = 0.052) from recessive models and 1.43 (1.03–1.99; number of studies = 3; $I^2 = 89.3\%$, p-heterogeneity < 0.001) from unspecified models. These altogether suggested that an effect of *MTHRF* of diabetes risk in African populations, if any, is weak and inconsistent across studies and genetic models. The pooled effects of the *KCNQ1*, *KCNJ11* and the *ENPP1* genes were non-significant regardless of the genetic models, with little evidence of statistical heterogeneity; suggesting that the contribution of these genes to diabetes occurrence in African populations is either trivial or uncertain based on available studies. Studies from sub-Saharan Africa largely reported estimates from unspecified genetic models, while studies from Northern Africa largely reported estimates from specific genetic model; precluding meaningful regional comparison of the pooled effect estimates. With the exception of dominant genetic models ($p = 0.025$), there was no evidence of publication bias as indicated in appendix pp. 22 (all $p > 0.262$ for the Egger's test of bias).

4. Discussion

Our review provides a comprehensive and integrated perspective of replication studies on the genetic of T2D in populations within the African continent. Extant studies have tested several polymorphisms in 57 genes reported to be associated with T2D in other populations. The included studies largely used a candidate gene approach, with a trend mostly toward consistent effects for genes tested in multiple studies, with a modest effect size of the variants with a positive signal. Regarding the effects of common variants with the greatest effects on the risk of type 2 diabetes in other populations, namely *TCF7L2* in Europeans, and KCNQ in Asians [34], their role in Africans is unclear. The two genome-wide studies revealed potential linkage signals in chromosomes regions harboring (or next to) these genes, while studies of selected SNPs of the two genes mostly found no association of KCNQ with diabetes risk.

The importance of performing genetic studies of diabetes in African populations to fine map the causal genetic variants was recently been emphasized by the work of the MEta-analysis of type 2 diabetes in African Americans (MEDIA) Consortium [35]. Although the MEDIA Consortium found that a substantial number of previously reported susceptibility loci for T2D were transferable to African Americans, they also identified novel susceptibility loci specific to this population. Furthermore, for those loci shared between African Americans and other populations, the strongest association in African Americans was often found in nearby SNPs and not necessarily the SNPs originally reported in other populations [35]. Therefore, just like trans-ethnic fine mapping, resequencing efforts are also needed to uncover novel signal and functional coding alleles in transcripts close to loci known to be causally related with T2D in other populations [36,37].

In line with fi dings of the MEDIA Consortium [35], meta-analyses of GWAS have consistently demonstrated that some genetic diabetes risk signals were transferrable across populations, while others were population specifi [6,38]. Our review found only two studies that had examined the human genome using GWAS in the same African population, but which were underpowered to possibly uncover those diabetes risk loci that are specific to populations within Africa [15,16]. Of the genetic variants reported to confer the strongest risk of diabetes in Europeans (*TCF7L2*) and Asians (KCNQ) only *TCF7L2* was transferable to populations within Africa. Studies on the heterogeneity of the effects of transferrable genetic risk variants across population have been mostly consistent in reporting the similarities of effects estimates [6,39]. For a potentially susceptible gene such as *TCF7L2* with a positive signal across a few studies in our review, the pooled odd ratio was essentially within the range of those reported in Asians and Caucasians [6] suggesting a homogeneity of the genetic effects across population. This homogeneity if confirmed, would tend to project a prominent contribution of differences in environmental factors, to the currently observed variability in diabetes fi ures across populations.

The interest in studying the genetics of diabetes is driven among others by the prospects of translating the genetic findings to inform medical practice. Thus far the major contribution of genetic discoveries to medical practice regarding diabetes mellitus, has been limited to the tiny proportion of patients with rare monogenic forms of the disease [40]. There are suggestions that in these rare forms, the implicated risk alleles tend to affect the coding sequences of the DNA, resulting in predictable effects on the functions of the genes. This knowledge in turn can be used to improve the prognosis and treatment of the disease. In the polygenetic forms of the disease however, susceptibility signals are mostly located outside of the coding sequences the DNA, and may affect more regulation of the transcription and not the functions of the genes [41]. This makes very challenging the translation efforts of genetic information relating to the large majority of diabetes. However, possible applications include uncovering novel prevention and treatment targets, improving disease risk prediction and diagnosis, and personalized medicine [34]. With regard to prediction, available evidence suggests that common known alleles associated with T2D add little to disease prediction beyond knowledge from traditional diabetes risk factors [42,43]. Interest in investigating further the contribution of genetic information to diabetes prediction and diagnosis in Africa, is motivated by the existence of variants of likely type 2 diabetes with particular phenotypes, treatment and motoring requirements, that are more frequent in African population, yet regularly underdiagnosed in routine clinical setting [44].

We did not address the overlap of genes associated with T2D with other cardiometabolic traits or long-term complications of the disease, nor did we address monogenic diabetes. Some SNPs known to be associated with obesity, related traits and other cardiometabolic risk factors have been found to increase the risk of type 2 diabetes. Likewise, loci infl ncing type 2 diabetes risk, glycemic traits and insulin levels have been reported to be associated with obesity related traits and cardiovascular diseases risk [45–48]. In the presence of overlapping associations, uncriti- cal adjustment for the effect of other cardiometabolic factors on type 2 diabetes risk could obscure or attenuate the effects the tested loci [47].

The strengths of our review are manifold. We comprehensively and rigorously appraised the extant data, including the whole spectrum of evidence from African countries, which was not the case with previous reviews, and addressing issues specific and relevant to the African context and also identifying gaps in research. More importantly, we conducted meta-analyses increasing the likelihood of detecting the effects of variant examined across multiple studies. Our study does have limitations. First, existing studies (genetic association studies) did not always report key methodological information that include testing the HWE, the sample size/power calculations, clear description of controls, consideration and correction for population stratification, as well as the levels of adjustment conducted. Second, compared to studies conducted elsewhere, sample sizes were less substantial. It is possible that with larger sample sizes and more comprehensive platforms, some additional previously proposed candidate genes may reach statistical significance. The limited sample size indicates that the studied populations may not cover all the scope of human genetic variations that exist on the African continent. Third, some variants were investigated in single studies thus precluding a literature-based meta-analysis. Furthermore, we had no access to individual data precluding refined analyses and adjustment for potential confounders and other types of bias. Hence, we could not assess potential sources of bias.

Over the last ten years, there has been progress in the investigation of the genetic architecture of type 2 diabetes among African populations. However, important gaps are yet to be filled. Existing studies have mainly been confined to certain populations, thus precluding the full examination of the potential genetic variation across the continent. Hitherto, the conducted studies have been replication endeavor, thus it is still unclear whether there may be genetic variants that confer the risk of diabetes exclusively among African populations. Carefully and well-designed larger-scale studies are therefore needed, in which assessment of gene environment interaction can also be addressed. However, this would require well-coordinated multi-centered groups working within research consortia to establish large enough cohorts with recruitment in several African countries, especially since genetic research on the continent has been hindered by a number of challenges such as limited funding, as well as the lack of infrastructure and local expertise [49,50] Such coordinated efforts would enable going beyond the scope of the extant data, and conduct studies including family-based, gene–environment and gene–gene interactions, and genome-wide association analyses. Such studies may provide further insights in the etiology of type 2 diabetes.

In summary, we have presented a systematic and comprehensive assessment of the current evidence on genetic determinants of type 2 diabetes in Africa. The putative genetic risk factors that have emerged from current overview represent the most promising type 2 diabetes susceptibility genes described to date in Africans. However, larger-scale genetic studies will further expand our knowledge of the underlying genetic mechanisms of type 2 diabetes among Africans.

Author contribution

Y.Y.Y.: Acquired data; analysed data; interpreted data; drafted and revised the manuscript for intellectual content.

M.G.F: Acquired data; analysed data; interpreted data; drafted and revised the manuscript for intellectual content.

E.V.B.: Analysed data and interpreted data; and revised the manuscript for intellectual content.

N.B.N.: Designed the study, Interpreted data and revised the manuscript for intellectual content

T. E. M.: Revised the manuscript for intellectual content.

E.S.: Revised the manuscript for intellectual content.

R. T. E.: Revised the manuscript for intellectual content.

J.B.E: Conceived and designed the study, acquired the data, drafted and revised the manuscript for intellectual content.

A.P.K.: Conceived and designed the study; analysed data; interpreted data; drafted the manuscript; supervised the study and revised the manuscript for intellectual content.

All authors had full access to the data.

A.P.K is the guarantor.

Conflict of interest

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.diabres.2016.01.003.